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In supporting: SUSTAINABLE DEVELOPMENT GOALS

DESIGN AND DEVELOPMENT OF A SMART AGRICULTURE MONITORING PLATFORM FOR SUSTAINABLE FARMING PRACTICES

SURIANA BINTI ABDULLAH | ID: 012025021263 | BACHELOR IN COMPUTER SCIENCE

Smart agriculture prototype for mustard greens monitoring, irrigation automation and AI-assisted fertilizer decisions.

6+ Live readings

2 Pump controls

AI Fertilizer advice

PWA Mobile access

INTRODUCTION

- This project develops a smart agriculture platform for mustard greens monitoring.
- The system combines IoT sensors, Firebase, actuator control and AI recommendation.

PROBLEM STATEMENT

- Manual watering may delay irrigation response.
- Fertilizer decisions can rely on estimation without sensor-based support.
- Important crop and pump events may be missed without alerts.

OBJECTIVES

- Develop IoT irrigation control using ESP32 and Firebase.
- Create AI fertilizer recommendation based on selected sensor data.
- Provide notification alerts for important system events.

PROJECT SCOPES

- Mustard greens monitoring using soil, climate, TDS and water flow readings.
- React/Vite dashboard with live readings, trend view, controls and notifications.
- Prototype testing covers sensor updates, pump response, AI output and usability.

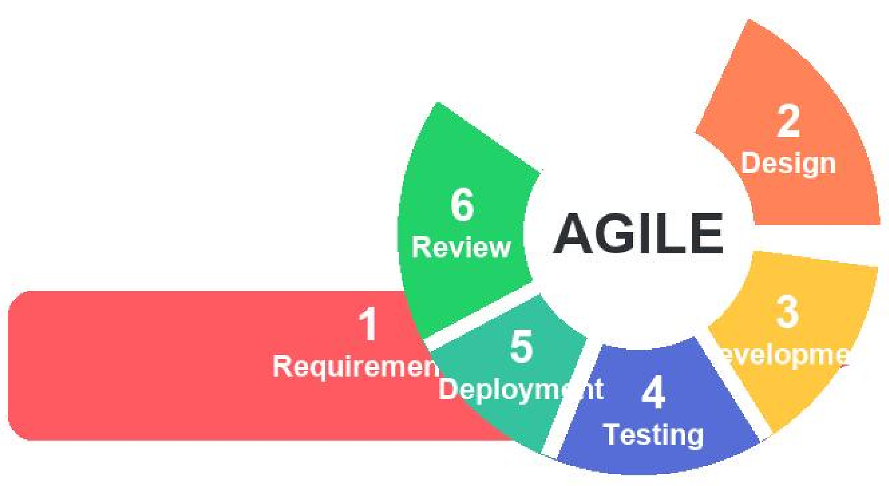
KEY MODULES

Sensor Reading | Firebase Sync | Auto Irrigation | AI Fertilizer | Trend Analysis | Notifications

DATA FLOW

Sensors → ESP32 → Firebase → React App → Pumps → Alerts

METHODOLOGY



Working Prototype
ESP32, sensors, relays, Firebase and mobile web app were integrated.

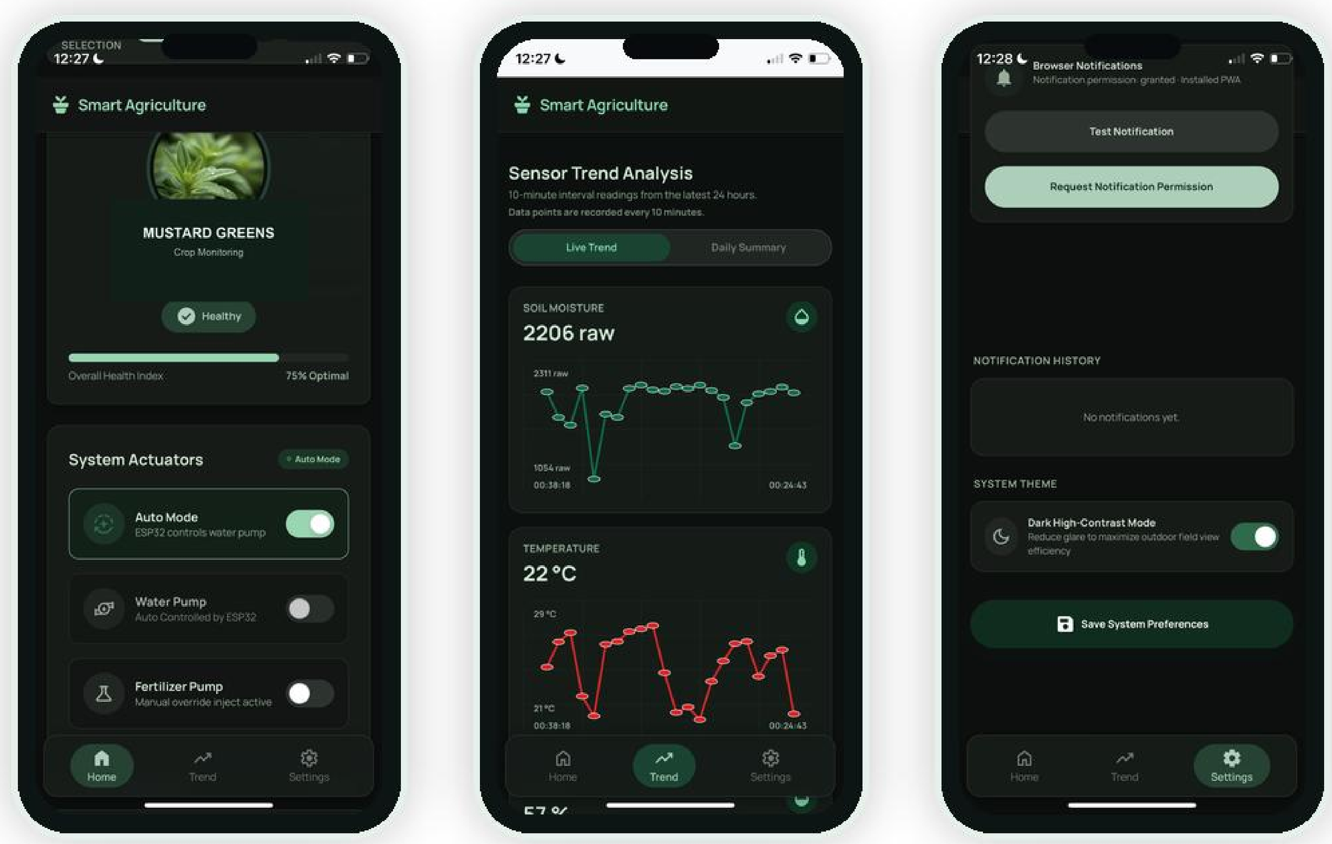
Smart Irrigation
Water pump responds to calibrated Wet, Normal and Dry soil conditions.

AI Decision Support
AI card explains fertilizer need with confidence output.

Sustainable Use
Water usage and alerts support mustard greens care.

RESULT

Application interface and tested output screens



Home | Trend | Alerts

CONCLUSION

The system combines monitoring, automation, cloud synchronization and AI recommendation to support sustainable smart agriculture for mustard greens.

SDG ALIGNMENT: SDG 2 Zero Hunger

Main agriculture image: Hydroponic farming, USA by Preston Keres/USDA/FPAC, public domain via Wikimedia Commons.

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